

```

1      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
NOTE: ODS statements in the SAS Studio environment may disable some output features.
69
70      /* CMP-135 SDTM EX Creation
71      Domain: EX*/;
72      options validvarname=upcase nofmterr dlcreatedir;
73
74      libname raw "/home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1";
NOTE: Libref RAW was successfully assigned as follows:
      Engine:          V9
      Physical Name: /home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1
75      libname sdtm "/home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm";
NOTE: Libref SDTM was successfully assigned as follows:
      Engine:          V9
      Physical Name: /home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm
76
77      libname valxpt xport
78      "/home/u64124548/sasuser.v94/SDTM - Creation/SDTM Data For Validation V1/ex.xpt";
NOTE: Libref VALXPT was successfully assigned as follows:
      Engine:          XPORT
      Physical Name: /home/u64124548/sasuser.v94/SDTM - Creation/SDTM Data For Validation V1/ex.xpt
79
80      libname sexpt xport
81      "/home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1/sdtm to use for vs ds/se.xpt";
NOTE: Libref SEXPT was successfully assigned as follows:
      Engine:          XPORT
      Physical Name: /home/u64124548/sasuser.v94/SDTM - Creation/crfdatav1/sdtm to use for vs ds/se.xpt
82
83
84      data se_prep;
85          set sexpt.se;
86
87          sestdt = input(substr(SESTDTC, 1, 10), yymmdd10.);
88          seenddt = input(substr(SEENDTC, 1, 10), yymmdd10.);
89
90          format sestdt seenddt date9.;
91
92          keep USUBJID EPOCH sestdt seenddt;
93      run;

NOTE: There were 85 observations read from the data set SEXPT.SE.
NOTE: The data set WORK.SE_PREP has 85 observations and 4 variables.
NOTE: DATA statement used (Total process time):
      real time          0.00 seconds
      user cpu time      0.01 seconds
      system cpu time    0.01 seconds
      memory             894.31k
      OS Memory          24484.00k
      Timestamp          06/29/2026 07:28:58 PM
      Step Count         480   Switch Count   2
      Page Faults        0
      Page Reclaims     116
      Page Swaps         0
      Voluntary Context Switches 19
      Involuntary Context Switches 0
      Block Input Operations 0
      Block Output Operations 264

94
95
96
97      data ex_pre;
98          length
99          STUDYID $6

```

```

100          DOMAIN    $2
101          EXTRT      $10
102          EXDOSU     $2
103          EXDOSFRM   $9
104          EXROUTE    $12
105          EXLOC2     $20
106          EXSTDTC    $16
107          USUBJID    $40;
108
109          set raw.ex;
110
111          STUDYID = "CMP135";
112          DOMAIN  = "EX";
113          USUBJID = catx("-", STUDYID, strip(SUBJECT));
114
115          EXTRT    = "CMP135";
116          EXDOSE2  = EXDOSE;
117          EXDOSU   = "mg";
118          EXDOSFRM = "INJECTION";
119          EXROUTE  = "SUBCUTANEOUS";
120
121          if not missing(EXLOCOTH) then EXLOC2 = upcase(strip(EXLOCOTH));
122          else EXLOC2 = upcase(strip(EXLOC));
123
124          if vtype(EXSTDN) = "N" then do;
125              if not missing(EXSTDN) then do;
126                  if abs(EXSTDN) > 50000 then EXSTDTC = datepart(EXSTDN);
127                  else EXSTDTC = EXSTDN;
128              end;
129          end;
130          else do;
131              EXDTM_RAW = input(strip(EXSTDN), anydtdtm.);
132              if not missing(EXDTM_RAW) then EXSTDTC = datepart(EXDTM_RAW);
133              if missing(EXSTDTC) then EXSTDTC = input(strip(EXSTDN), anydtdte.);
134          end;
135
136          if vtype(EXSTTM) = "N" then do;
137              if EXSTTM > 86400 then EXSTM = timepart(EXSTTM);
138              else EXSTM = EXSTTM;
139          end;
140          else EXSTM = input(strip(EXSTTM), time.);
141
142          if not missing(EXSTDTC) and not missing(EXSTM) then
143              EXSTDTC = cats(put(EXSTDTC, yymmdd10.), "T", put(EXSTM, tod5.));
144          else if not missing(EXSTDTC) then
145              EXSTDTC = put(EXSTDTC, yymmdd10.);
146
147          format EXSTDTC date9. EXSTM time8.;
148
149          keep STUDYID DOMAIN USUBJID EXTRT EXDOSE2 EXDOSU EXDOSFRM
150              EXROUTE EXLOC2 EXSTDTC EXSTDTC SUBJECT RECORDID RECORDPOSITION;
151          run;

```

NOTE: Numeric values have been converted to character values at the places given by: (Line):(Column).
131:33 133:54

NOTE: Character values have been converted to numeric values at the places given by: (Line):(Column).
137:12 137:49 138:22

NOTE: There were 1904 observations read from the data set RAW.EX.

NOTE: The data set WORK.EX_PRE has 1904 observations and 14 variables.

NOTE: DATA statement used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	2002.34k
OS Memory	25256.00k
Timestamp	06/29/2026 07:28:58 PM

Step Count	481	Switch Count	2
Page Faults	0		
Page Reclaims	195		
Page Swaps	0		
Voluntary Context Switches	18		
Involuntary Context Switches	0		
Block Input Operations	0		
Block Output Operations	784		

152
153

```
154      proc sql;
155          create table ex_rfstdt as
156          select
157              SUBJECT,
158              min(EXSTDT) as RFSTDT format=date9.
159          from ex_pre
160          where not missing(EXSTDT)
161          group by SUBJECT;
```

NOTE: Table WORK.EX_RFSTDТ created, with 24 rows and 2 columns.

162 quit;

NOTE: PROCEDURE SQL used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	6021.90k
OS Memory	29992.00k
Timestamp	06/29/2026 07:28:58 PM
Step Count	482
Switch Count	5
Page Faults	0
Page Reclaims	196
Page Swaps	0
Voluntary Context Switches	17
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	272

163
164

```
165      proc sql;
166          create table ex_with_dy as
167          select
168              a.*,
169              b.RFSTDT
170          from ex_pre as a
171          left join ex_rfstdt as b
172              on a.SUBJECT = b.SUBJECT;
```

NOTE: Table WORK.EX_WITH_DY created, with 1904 rows and 15 columns.

173 quit;

NOTE: PROCEDURE SQL used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	11523.46k
OS Memory	34272.00k
Timestamp	06/29/2026 07:28:58 PM
Step Count	483
Switch Count	2
Page Faults	0
Page Reclaims	479
Page Swaps	0
Voluntary Context Switches	22
Involuntary Context Switches	0

Block Input Operations	0
Block Output Operations	784

```

174
175     data ex_with_dy;
176         set ex_with_dy;
177
178         if not missing(EXSTDT) and not missing(RFSTDT) then do;
179             if EXSTDT >= RFSTDT then EXSTDY = EXSTDT - RFSTDT + 1;
180             else EXSTDY = EXSTDT - RFSTDT;
181         end;
182     run;

```

NOTE: There were 1904 observations read from the data set WORK.EX_WITH_DY.

NOTE: The data set WORK.EX_WITH_DY has 1904 observations and 16 variables.

NOTE: DATA statement used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.01 seconds
memory	1701.96k
OS Memory	25896.00k
Timestamp	06/29/2026 07:28:58 PM
Step Count	484
Page Faults	0
Page Reclaims	123
Page Swaps	0
Voluntary Context Switches	11
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	776

```

183
184
185     proc sql;
186         create table ex_epoch_match as
187         select
188             a.*,
189             b.EPOCH,
190             b.SEESTDT,
191             b.SEENDT
192         from ex_with_dy as a
193         left join se_prep as b
194             on a.USUBJID = b.USUBJID
195             and a.EXSTDT >= b.SEESTDT
196             and a.EXSTDT <= b.SEENDT;

```

NOTE: Table WORK.EX_EPOCH_MATCH created, with 1962 rows and 19 columns.

```

197     quit;
NOTE: PROCEDURE SQL used (Total process time):
real time      0.00 seconds
user cpu time   0.01 seconds
system cpu time 0.00 seconds
memory         8468.09k
OS Memory      31200.00k
Timestamp      06/29/2026 07:28:58 PM
Step Count     485
Page Faults    0
Page Reclaims  429
Page Swaps     0
Voluntary Context Switches 23
Involuntary Context Switches 2
Block Input Operations 0
Block Output Operations 1048

```

```

198
199     proc sort data=ex_epoch_match;
200         by STUDYID USUBJID RECORDID RECORDPOSITION descending SESTD;
201     run;

```

NOTE: There were 1962 observations read from the data set WORK.EX_EPOCH_MATCH.

NOTE: The data set WORK.EX_EPOCH_MATCH has 1962 observations and 19 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	2553.12k
OS Memory	27060.00k
Timestamp	06/29/2026 07:28:58 PM
Step Count	486
Page Faults	0
Page Reclaims	263
Page Swaps	0
Voluntary Context Switches	10
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	1032

```

202
203     data ex_with_epoch;
204         set ex_epoch_match;
205         by STUDYID USUBJID RECORDID RECORDPOSITION;
206
207         if first.RECORDPOSITION;
208
209         drop SESTD SEENDT RFSTD;
210     run;

```

NOTE: There were 1962 observations read from the data set WORK.EX_EPOCH_MATCH.

NOTE: The data set WORK.EX_WITH_EPOCH has 1904 observations and 16 variables.

NOTE: DATA statement used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	1917.81k
OS Memory	26024.00k
Timestamp	06/29/2026 07:28:58 PM
Step Count	487
Page Faults	0
Page Reclaims	155
Page Swaps	0
Voluntary Context Switches	12
Involuntary Context Switches	2
Block Input Operations	0
Block Output Operations	776

```

211
212
213     proc sort data=ex_with_epoch out=ex_sorted;
214         by STUDYID USUBJID EXSTD EXSTDC RECORDPOSITION;
215     run;

```

NOTE: There were 1904 observations read from the data set WORK.EX_WITH_EPOCH.

NOTE: The data set WORK.EX_SORTED has 1904 observations and 16 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds

```

memory                2687.93k
OS Memory              27192.00k
Timestamp              06/29/2026 07:28:58 PM
Step Count             488   Switch Count   2
Page Faults            0
Page Reclaims          249
Page Swaps             0
Voluntary Context Switches 11
Involuntary Context Switches 0
Block Input Operations  0
Block Output Operations 784

```

```

216
217     data sdtm.ex;
218         retain
219             STUDYID DOMAIN EXSEQ EXTRT EXDOSE EXDOSU EXDOSFRM
220             EXROUTE EXLOC EPOCH EXSTDTC EXSTDY USUBJID;
221
222         length
223             STUDYID  $6
224             DOMAIN   $2
225             EXTRT    $10
226             EXDOSU   $2
227             EXDOSFRM $9
228             EXROUTE  $12
229             EXLOC    $20
230             EPOCH    $11
231             EXSTDTC  $16
232             USUBJID  $40;
233
234         set ex_sorted;
235         by STUDYID USUBJID;
236
237         if first.USUBJID then EXSEQ = 1;
238         else EXSEQ + 1;
239
240         EXDOSE = EXDOSE2;
241         EXLOC  = EXLOC2;
242
243         label
244             STUDYID  = "Study Identifier"
245             DOMAIN   = "Domain Abbreviation"
246             EXSEQ    = "Sequence Number"
247             EXTRT    = "Name of Actual Treatment"
248             EXDOSE   = "Dose per Administration"
249             EXDOSU   = "Dose Units"
250             EXDOSFRM = "Dose Form"
251             EXROUTE  = "Route of Administration"
252             EXLOC    = "Location of Dose Administration"
253             EPOCH    = "Epoch"
254             EXSTDTC  = "Start Date/Time of Treatment"
255             EXSTDY   = "Study Day of Start of Treatment";
256
257         keep
258             STUDYID DOMAIN EXSEQ EXTRT EXDOSE EXDOSU EXDOSFRM
259             EXROUTE EXLOC EPOCH EXSTDTC EXSTDY USUBJID;
260     run;

```

NOTE: There were 1904 observations read from the data set WORK.EX_SORTED.

NOTE: The data set SDTM.EX has 1904 observations and 13 variables.

NOTE: DATA statement used (Total process time):

```

real time          0.01 seconds
user cpu time      0.01 seconds
system cpu time    0.00 seconds
memory            1793.71k

```

OS Memory	25896.00k
Timestamp	06/29/2026 07:28:58 PM
Step Count	489
Page Faults	0
Page Reclaims	123
Page Swaps	0
Voluntary Context Switches	50
Involuntary Context Switches	1
Block Input Operations	0
Block Output Operations	776

261

262

263 proc sort data=sdtm.ex out=ex_created;

264 by STUDYID USUBJID EXSEQ;

265 run;

NOTE: There were 1904 observations read from the data set SDTM.EX.

NOTE: The data set WORK.EX_CREATED has 1904 observations and 13 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.00 seconds
memory	2596.75k
OS Memory	27192.00k
Timestamp	06/29/2026 07:28:58 PM
Step Count	490
Page Faults	0
Page Reclaims	223
Page Swaps	0
Voluntary Context Switches	23
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	784

266

267 proc sort data=valxpt.ex out=ex_validate;

268 by STUDYID USUBJID EXSEQ;

269 run;

NOTE: There were 1904 observations read from the data set VALXPT.EX.

NOTE: The data set WORK.EX_VALIDATE has 1904 observations and 13 variables.

NOTE: PROCEDURE SORT used (Total process time):

real time	0.01 seconds
user cpu time	0.01 seconds
system cpu time	0.00 seconds
memory	18783.51k
OS Memory	42532.00k
Timestamp	06/29/2026 07:28:59 PM
Step Count	491
Page Faults	0
Page Reclaims	1044
Page Swaps	0
Voluntary Context Switches	30
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	776

270

271 ods rtf file="/home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm/EX_PROC_COMPARE.rtf";

NOTE: Writing RTF Body file: /home/u64124548/sasuser.v94/SDTM - Creation/output_sdtm/EX_PROC_COMPARE.rtf

272

273 title "PROC COMPARE: SDTM EX Created vs Validation EX";

```
274
275     proc compare base=ex_validate compare=ex_created listall criterion=0.0000001;
276         id STUDYID USUBJID EXSEQ;
277     run;
```

NOTE: There were 1904 observations read from the data set WORK.EX_VALIDATE.

NOTE: There were 1904 observations read from the data set WORK.EX_CREATED.

NOTE: PROCEDURE COMPARE used (Total process time):

real time	0.02 seconds
user cpu time	0.02 seconds
system cpu time	0.01 seconds
memory	2775.34k
OS Memory	27308.00k
Timestamp	06/29/2026 07:28:59 PM
Step Count	492
Switch Count	0
Page Faults	0
Page Reclaims	388
Page Swaps	0
Voluntary Context Switches	0
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	16

```
278
279     ods rtf close;
280
281
282     title "Created EX - First 5 Records";
283     proc print data=ex_created(obs=5);
284 run;
```

NOTE: There were 5 observations read from the data set WORK.EX_CREATED.

NOTE: PROCEDURE PRINT used (Total process time):

real time	0.01 seconds
user cpu time	0.01 seconds
system cpu time	0.00 seconds
memory	1208.59k
OS Memory	26404.00k
Timestamp	06/29/2026 07:28:59 PM
Step Count	493
Switch Count	0
Page Faults	0
Page Reclaims	99
Page Swaps	0
Voluntary Context Switches	0
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	16

```
285
286     title "Validation EX - First 5 Records";
287     proc print data=ex_validate(obs=5);
288 run;
```

NOTE: There were 5 observations read from the data set WORK.EX_VALIDATE.

NOTE: PROCEDURE PRINT used (Total process time):

real time	0.01 seconds
user cpu time	0.02 seconds
system cpu time	0.00 seconds
memory	987.28k
OS Memory	26404.00k
Timestamp	06/29/2026 07:28:59 PM
Step Count	494
Switch Count	0
Page Faults	0
Page Reclaims	93

Page Swaps	0
Voluntary Context Switches	0
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	0

```
289
290     title;
291
292     OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
302
```